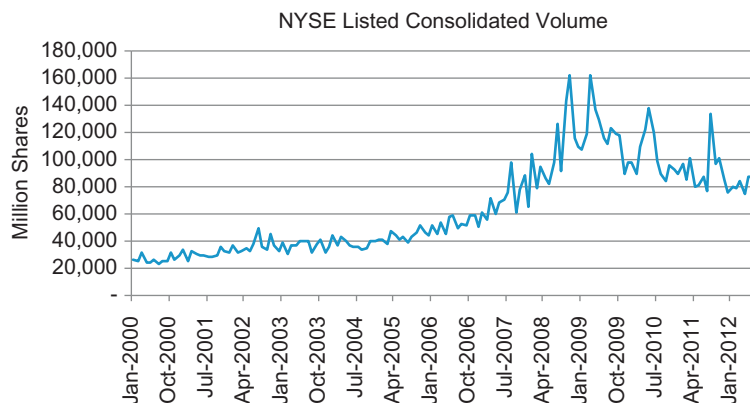


stable, trading difficulty, as measured by the number of transactions required to complete the order, has in fact increased.

Another major structural event that occurred with dramatic effects on trading was Reg NMS, regulation of national market systems. Reg NMS was a series of initiatives to modernize our national securities markets. The main goals were to promote competition across market centers and orders, and provide fair and efficient pricing mechanisms. The rules that had the greatest effect on the markets were:

- Order Protection (Rule 611)—Establish, maintain, and enforce written policies and procedures to prevent “trade-throughs.” Trade-throughs are the execution of trades at prices inferior to protected quotes. A protected quote is an order that is immediately and automatically accessible.
- Access (Rule 610)—Requires fair and non-discriminatory access to quotes and routing of orders to those market centers with the best prices.
- Sub Penny Pricing (Rule 612)—minimum pricing increments.

Analysis of market volumes has found that while visible liquidity at the best bid and ask may have decreased, actual traded volume has increased dramatically. Figure 1.1 shows the increase in volume for NYSE listed securities from 2000 to 2012. Total consolidated volume peaked during the financial crisis in 2008–2009 where it was more than $6\times$ the average in 2000. Volume has succumbed since 2009, but in 2012 it is still $3.3\times$ the 2000 average. While volumes increased in the market, the issue of fragmentation transpired. Traded volume in the NYSE was between 80 and 90% on average for 2000 through 2005, but then started decreasing



■ Figure 1.1 NYSE Listed Consolidated Volume.